SUPPLEMENT

1. Search Strategy

EBM Reviews:

((incidence* or epidemio*).ab,fs,hw,ti or (((number or new) adj2 cases) or surveillance or biosurveillance or pharmacoepidemiolog* or seroepidemio* or (morbidity adj2 (rate* or pattern*)) or infection-rate* or sampl* or age-distribution or se-distribution or geographic-distribution* or case-mi or community-assessment* or disease-registr* or prevalence).ab,fs,hw,ti.) AND (endocarditis.hw,ti. or (endocarditis adj5 (incidence* or epidemio* or cases or surveillance or biosurveillance or pharmacoepidemiolog* or seroepidemio* or rate* or prevalence)).ab,hw,ti.) NOT (“case report*”.pt,ti.) Limit to English, 2000+

Embase (1974+)

(ep incidence/ or ep epidemiological data/ or ep epidemiology/ or (((number or new) adj2 cases) or surveillance or biosurveillance or pharmacoepidemiolog* or seroepidemio* or (morbidity adj2 (rate* or pattern*)) or infection-rate* or sampl* or age-distribution or se-distribution or geographic-distribution* or case-mi or community-assessment* or disease-registr* or prevalence).ab,kw,ti.) AND (endocarditis/ or bacterial endocarditis/ or fungal endocarditis/ or prosthetic valve endocarditis/ or subacute endocarditis/ or endocarditis.kw,ti. or (endocarditis adj5 (incidence* or epidemio* or cases or surveillance or biosurveillance or pharmacoepidemiolog* or seroepidemio* or rate* or prevalence)).ab.) NOT (ep animal/ not ep human/, 12 "case report".kw,pt,ti.) Limit to English, 2000+, article or article in press

Ovid MEDLINE(R) 1946 to Present and Epub Ahead of Print, In-Process & Other Non-Indeed Citations and Ovid MEDLINE(R) Daily:

(Incidence/ or ep Epidemiology/ or ep Population Surveillance/ or (incidence* or epidemio*).ab,hw,kf,ti,f. or (((number or new) adj2 cases) or surveillance or biosurveillance or pharmacoepidemiolog* or seroepidemio* or (morbidity adj2 (rate* or pattern*)) or infection-rate* or sampl* or age-distribution or se-distribution or geographic-distribution* or case-mi or community-assessment* or disease-registr* or prevalence).ab,kf,ti.) AND (Endocarditis, Bacterial/ or Endocarditis/ or Endocarditis, Subacute Bacterial/ or endocarditis.kw,ti. or (endocarditis adj5 (incidence* or epidemio* or cases or surveillance or biosurveillance or
pharmacoepidemiolog* or seroepidemio* or rate* or prevalence)).ab.) NOT (ep Animals/ not Humans/, "case report".kw,pt,ti.) Limit to English, 2000+

Scopus:

(((TITLE (endocarditis) OR KEY (endocarditis) AND TITLE-ABS-KEY (endocarditis W/5 (
incidence* OR epidemio* OR cases OR surveillance OR biosurveillance OR pharmacoepidemiolog* OR seroepidemio* OR rate* OR prevalence)))
) AND ((TITLE (endocarditis) OR KEY (endocarditis) AND TITLE-ABS-KEY (endocarditis W/5 (
incidence* OR epidemio* OR cases OR surveillance OR biosurveillance OR pharmacoepidemiolog* OR seroepidemio* OR rate* OR prevalence))))
) AND NOT (TITLE ("case report*" OR animal* OR rat OR rats OR mice OR mouse OR rabbit* OR rodent*)) AND (LIMIT-TO (DOCTYPE
, "ar") OR LIMIT-TO (DOCTYPE , "re") OR LIMIT-TO (DOCTYPE , "er")) AND (LIMIT-TO (LANGUAGE , "English")) AND (LIMIT-TO (
SRCTYPE , "j")) AND (LIMIT-TO (PUBYEAR , 2020) OR LIMIT-TO (PUBYEAR , 2019) OR LIMIT-TO (PUBYEAR , 2018) OR LIMIT-TO (
PUBYEAR , 2017) OR LIMIT-TO (PUBYEAR , 2016) OR LIMIT-TO (PUBYEAR , 2015) OR LIMIT-TO (PUBYEAR , 2014) OR LIMIT-TO (
PUBYEAR , 2013) OR LIMIT-TO (PUBYEAR , 2012) OR LIMIT-TO (PUBYEAR , 2011) OR LIMIT-TO (PUBYEAR , 2010) OR LIMIT-TO (
PUBYEAR , 2009) OR LIMIT-TO (PUBYEAR , 2008) OR LIMIT-TO (PUBYEAR , 2007) OR LIMIT-TO (PUBYEAR , 2006) OR LIMIT-TO (

PUBYEAR , 2005) OR LIMIT-TO (PUBYEAR , 2004) OR LIMIT-TO (PUBYEAR , 2003) OR LIMIT-TO (PUBYEAR , 2002) OR LIMIT-TO (PUBYEAR , 2001) OR LIMIT-TO (PUBYEAR , 2000))

Web of Science:

(TOPIC: (incidence* or epidemio*) *OR* TOPIC: ((((number or new) NEAR/2 cases) or surveillance or biosurveillance or pharmacoepidemiolog* or seroepidemio* or (morbidity NEAR/2 (rate* or pattern*)) or infection-rate* or sampl* or age-distribution or se-distribution or geographic-distribution* or case-mi or community-assessment* or disease-registr* or prevalence)) AND TOPIC: (endocarditis) *OR* TOPIC: (endocarditis NEAR/5 (incidence* or epidemio* or cases or surveillance or biosurveillance or pharmacoepidemiolog* or seroepidemio* or rate* or prevalence))) NOT TI=("case report*" OR animal* OR rat OR rats OR mice OR mouse OR rabbit* OR rodent*), Limit to English, Article, 2000+.

Table 2. ICD 10 code designations (green highlight = included).

	Studies				
ICD-10 Codes	**Alkhouli 2019**	**Kadri 2019**	**McCarthy 2020**	**Wong 2020**	**Garg 2019**
I33.0					
I39					
I33.9					
I38					
A39.51					
A54.83					
B37.6					
B39.4					
B39.5					
B39.9					
I33					

Table 3. ICD 9 code designations.

	Studies										
ICD-9 Codes	**Mendiratta 2009**	**Alkhouli 2019**	**Kadri 2019**	**Mori 2020**	**Bikdeli 2013**	**Toyoda 2017**	**Thornhill 2018**	**Moeryra 2019**	**McCarthy 2020**	**Wong 2020**	**Garg 2019**
424											
421.0											
421.1											
421.9											
112.81											
036.42											
098.84											
115.04											
115.14											
115.94											
424.9											
424.91											
424.99											

424.90											
421											

Table 4. Primary and Secondary diagnosis code designations.

Studies	Year	Primary + Secondary codes
Tleyjeh	2005	N/A
Mendiratta	2009	P+S
Correa	2010	N/A
Bikdeli	2013	P+S
DeSimone	2015	N/A
Toyoda	2017	N/A
Thornhill	2018	P+S
Alkhouli	2019	P+S
Garg	2019	N/A
Kadri	2019	P+S
Moeryra	2019	P+S
McCarthy	2020	N/A
Mori	2020	N/A
Wong	2020	N/A

N/A = non-applicable.

Table 5. Quality Assessment of included studies.

5.1. Adequacy of population definition

Study	Adequacy of population definition	Comment on age criteria
Tleyjeh 2005	Adequate	all adults (≥18 years) residing in Olmsted county, Minnesota, USA
Mendiratta 2009	Adequate	Patients 65 and older
Correa 2010	Adequate	all adults (≥18 years) residing in Olmsted county, Minnesota, USA
Bikdeli 2013	Adequate	Patients 65 and older
deSimone 2015	Adequate	all adults (≥18 years) residing in Olmsted county, Minnesota, USA
Thornhill 2018	Adequate	Patients older than 18
Toyoda 2017	Adequate	Patients 18 and older
Alkhouli 2019	Adequate	Patients older than 18
Garg 2019	Adequate	Patients aged 18-104
Kadri 2019	Adequate	Patients 18 and older
Moreyra 2019	Adequate	Patients older than 18
McCarthy 2020	Adequate	Patients 18 and older
Mori 2020	Adequate	Patients 18 and older
Wong 2020	Adequate	Patients aged 18-64
Population definition was deemed to be inadequate if the residency status of all IE patients was not ascertained. Definition of the population, which includes geographic boundaries with verification of residency status, temporal boundaries, or other boundaries, such as restricting cases to adults or to native-valve IE.		

5.2. Sampling Techniques

Study	Sampling techniques	Comment
Tleyjeh 2005	Adequate	Complete enumeration
Mendiratta 2009	Adequate	Complete enumeration
Correa 2010	Adequate	Complete enumeration
Bikdeli 2013	Adequate	Complete enumeration
deSimone 2015	Adequate	Complete enumeration
Toyoda 2017	Adequate	Complete enumeration
Thornhill 2018	Adequate	Complete enumeration
Alkhouli 2019	Adequate	Complete enumeration
Garg 2019	Adequate	Complete enumeration
Kadri 2019	Adequate	Complete enumeration
Moreyra 2019	Adequate	Complete enumeration
McCarthy 2020	Adequate	Complete enumeration
Mori 2020	Adequate	Complete enumeration
Wong 2020	Adequate	Complete enumeration
Optimal sampling techniques include complete enumeration or random sampling techniques. Sampling list and sampling techniques (convenience sample versus complete enumeration vs probability sample).		

5.3. Disease Definition

Study	Disease definition	Comment
Tleyjeh 2005	Adequate	Modified duke. Definite or possible.
Mendiratta 2009	Inadequate	ICD 9. Primary and secondary
Correa 2010	Adequate	Modified duke. Definite or possible.
Bikdeli 2013	Inadequate	ICD 9 and 10. Primary and secondary.
deSimone 2015	Adequate	Modified duke. Definite or possible.
Toyoda 2017	Inadequate	ICD 9 and 10. Primary and secondary.
Thornhill 2018	Inadequate	ICD 9. Primary and secondary
Alkhouli 2019	Inadequate	ICD 9 and 10. Primary and secondary
Garg 2019	Inadequate	ICD 9. Primary and secondary
Kadri 2019	Inadequate	ICD 9 and 10. Primary and secondary.
Moreyra 2019	Inadequate	ICD 9. Primary and secondary
McCarthy 2020	Inadequate	ICD 9 and 10. Primary
Mori 2020	Inadequate	ICD 9 and 10. Primary and secondary.
Wong 2020	Inadequate	ICD 9 and 10
For the purpose of our study, we did not consider a specific case definition to be superior because case definitions have changed over time with the advent of echocardiography and improvement in blood culture techniques. Definition of the disease (IE), which includes diagnostic criteria (Duke vs von Reyn or others) and levels of diagnostic certainty (definite, probable, and possible).		

5.4. Completeness of case ascertainment

Study	Completeness of case ascertainment	Comment
Tleyjeh 2005	Adequate	Independent validation
Mendiratta 2009	Adequate	Record linkage using ICD codes
Correa 2010	Adequate	Independent validation
Bikdeli 2013	Adequate	Record linkage using ICD codes
deSimone 2015	Adequate	Independent validation
Toyoda 2017	Adequate	Independent validation
Thornhill 2018	Adequate	Record linkage using ICD codes
Alkhouli 2019	Adequate	Record linkage using ICD codes
Garg 2019	Adequate	Independent validation
Kadri 2019	Adequate	Record linkage using ICD codes
Moreyra 2019	Adequate	Record linkage using ICD codes
McCarthy 2020	Adequate	Record linkage using ICD codes
Mori 2020	Adequate	Record linkage using ICD codes
Wong 2020	Adequate	Record linkage using ICD codes
Adequacy of case ascertainment was assessed on the basis of case finding procedures, inclusion of postmortem diagnoses, and number of hospitals serving the population under study that participated in the study. Case ascertainment that depends on the case finding procedures (census of existing diagnoses, record-linkage system, and direct-contact survey).		

6. Statistical evaluation of trends in prevalence of PWID among IE patients described in individual studies.

Studies	**P value**	**Annual Percentage Change % (95% Confidence Interval)**
Toyoda 2017†	< 0.001	0.9 (0.4-1.3)
Alkhouli 2019†	< 0.001	NR
Kadri 2019	< 0.001	1.8 (1.7-1.9)
McCarthy 2020	NR	NR
Mori 2020†	< 0.0001	NR
Wong 2020	< 0.01	5.7 (4.9-6.4)**

**NR: Not reported*

***Change in incidence of PWID in IE patients estimated using multi-variate Poisson regression analysis*

† Used the NIS database